

SafeSpeak: Real-Time Cyberbullying Detection Before Social Media Posting

Sujal Keer

Dept. of Data Science, Patkar Varde College

sujalkeer715@gmail.com



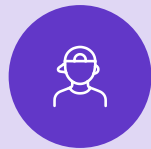
The Rationale

Why This Project Matters



The Cyberbullying Crisis

Defined as harassment or humiliation via digital messages, cyberbullying is pervasive, affecting mental health and digital citizenship.



Urgency on Social Media

Approximately 60% of teenagers report having been bullied or harassed online, illustrating a critical failure point in current platform design.



The Lasting Impact

The emotional damage is immediate and profound, often leaving victims vulnerable long after a harmful post is removed.

“One post can leave a scar
for life.”



Current Moderation Fails to Prevent Harm

Reactive Moderation

Current systems are designed to remove posts **after** the harm has occurred, focusing on content removal rather than primary prevention.

Victim Already Affected

By the time a harmful post is flagged, reviewed, and taken down, the victim has already been exposed to the negative, damaging message.

Need for Prevention

We must pivot from solely treating the symptoms (removing posts) to implementing solutions that interrupt the act of harm itself.

The Paradigm Shift

Stopping Cyberbullying Before It Starts



User Types Message

The message input is intercepted in real-time, pre-submission.



System Warns

If the message is predicted to be harmful, the user receives an instant warning.



Action Blocked

The post is held until the harmful elements are revised or removed.

This model of preemptive intervention, driven by our "Safe Keyboard," transforms content moderation from reactive cleanup to proactive defense.



Our Strategic Approach

From Data Collection to Deployment



Data Curation

Collected and rigorously cleaned **57,466 real social media comments** (from Twitter, YouTube, and public Kaggle sets) to build a robust training corpus.



Model Training

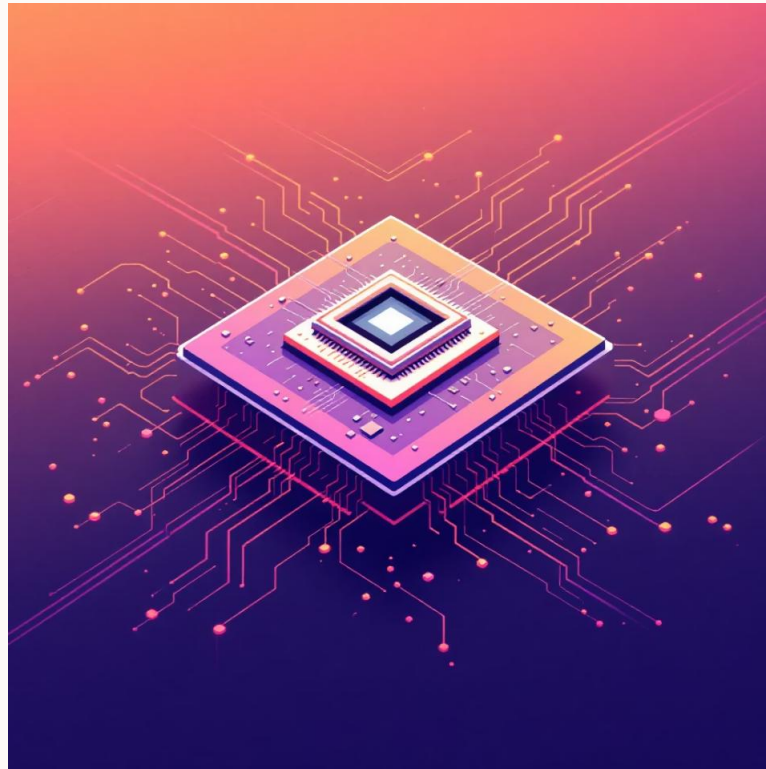
Developed and trained multiple sophisticated models, including classic machine learning (TF-IDF + ML) and advanced deep learning (BERT Transformer).



The "Safe Keyboard" Build

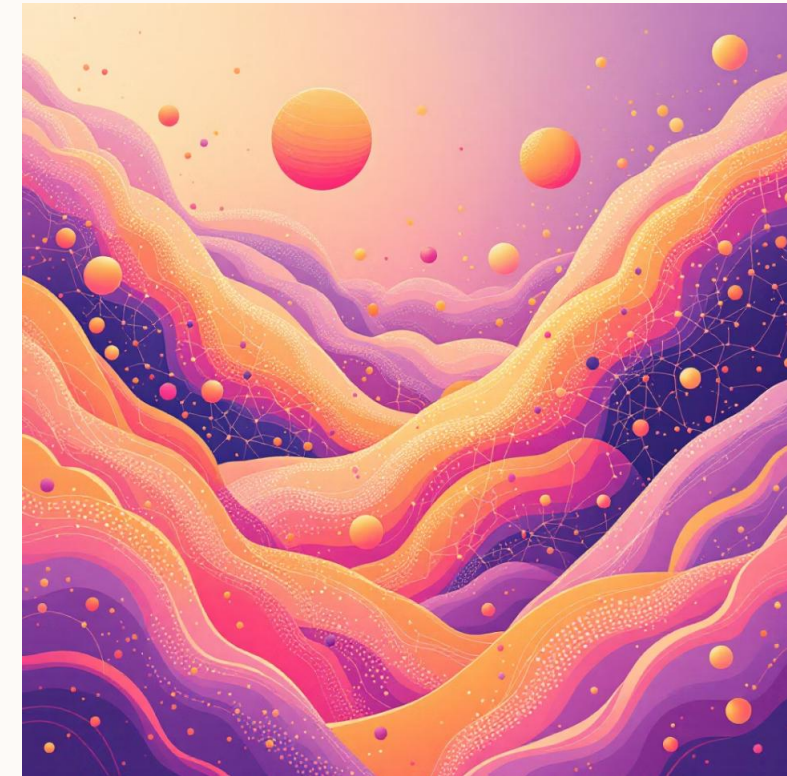
Created a functional prototype—the "Safe Keyboard"—capable of warning users instantly if their typed message is predicted to be harmful.

What Powers Our Real-Time Prediction



Classic Machine Learning

- Fast, lightweight, and efficient for high-speed checks.
- Good overall accuracy (72-80%) and **decisions in under 50 ms.**
- Ideal for platforms requiring minimal latency.



BERT Deep Learning Model

- State-of-the-art accuracy (84%) using contextual embeddings.
- Superior understanding of nuance, slang, and subtle bullying.
- The gold standard for comprehensive harm detection.

Accuracy and Speed of Core Models

Our models strike a critical balance between high accuracy (to catch subtle threats) and high speed (for a seamless user experience).

Model	Accuracy	Speed
BERT Transformer	84%	~200 ms
SVC	80%	~50 ms
Logistic Regression (TF-IDF)	78%	~50 ms
Multinomial Naive Bayes	72%	~50 ms

The Safe Keyboard in Action

Real-Time Pop-Up

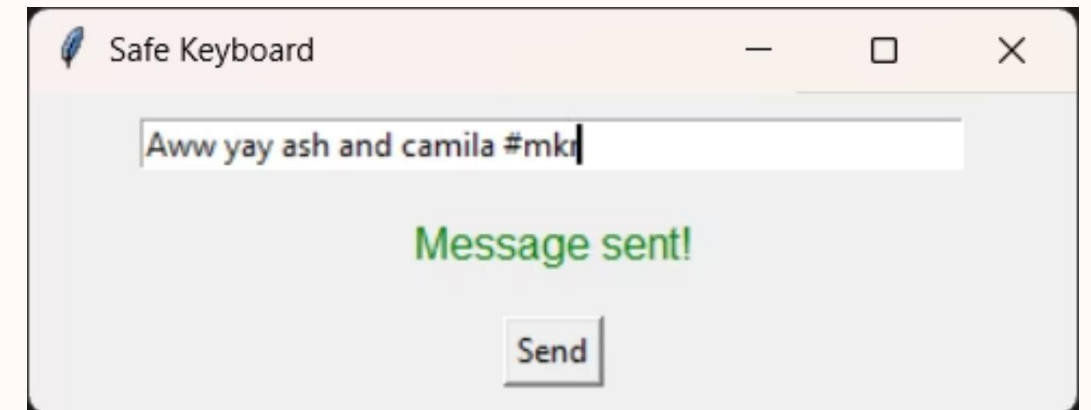
When harmful language is detected, an instant, non-intrusive notification appears, giving the user a chance to self-correct.

Mandatory Revision

"No message sent until cyberbullying is removed." This enforced pause is the core mechanism of prevention.

Visual Feedback

Red indicator means the post contains bullying or harassment;
Green means it's safe to send.



Scaling Protection and Improving Nuance



Language Expansion

Extend detection capabilities to new markets by adding support for non-English languages, starting with Hindi and Spanish.



Data Refinement

Invest in improving data quality to better handle slang, informal language, and nuanced context that often challenge current models.



Real-World Deployment

Conduct user studies and pilot the "Safe Keyboard" within smaller platforms to gather feedback and validate effectiveness in live environments.

Thank you